

Results of microwave thermographic profiles in 24 patients with biopsy-proven cancers may be summarized as follows:

1. Positive results were seen in 5 of 6 patients with lymphoma, in 7 of 8 women with recurrent breast carcinoma and in 3 of 5 women with primary breast cancer.
2. Negative results were obtained in 1 of 8 women with recurrent breast carcinoma, in 2 of 5 women with primary breast cancer, in 1 patient with multiple metastatic intramuscular foci of renal cell carcinoma and in 5 patients with more deeply-seated tumors: 2 lung, 1 esophagus, 1 hepatoma and 1 with lytic metastatic lesions in the femur and humerus.

The use of heat to enhance tumor image and other fundamental questions regarding the microwave detector are currently being investigated using the VX2 carcinoma in New Zealand rabbits. (This work was supported in part by the National Aeronautics and Space Administration Contract No. NAS1-15393)

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RADIOTHERAPY OF OCULAR MELANOMAS: PRECISION HIGH DOSE EXTERNAL BEAM PROTON THERAPY

Munzenrider, J.E.¹, Gragoudas, E.², Verhey, L.¹, Goitein, M.¹,
Koehler, A.³ and Suit, H.D.¹

Department of Radiation Medicine, Massachusetts General Hospital, Boston, MA¹
Department of Ophthalmology, Harvard Medical School, Boston, MA²
Harvard Cyclotron Laboratory, Cambridge, MA³

Fifty-two eyes in fifty-one patients with intra-ocular melanoma have received high dose fractionated precision proton beam therapy with a 160 MeV modulated proton beam. Twenty-nine male and 32 female patients ranged in age from 25-83 years. Diagnostic evaluation prior to treatment included complete ocular evaluation, fluorescein angiography, fundus photography and ultrasonography. Diameters of the tumor were measured directly on the sclera during surgical placement of 2 mm diameter tantalum rings around the perimeter of the tumor, which was determined by indirect ophthalmoscopy and transillumination; P32 uptake measurements were also taken at the time of clip placement. Tumor height was determined by immersion ultrasonography.

Tumors were grouped according to largest tumor diameter and tumor height with large, intermediate, and small tumors having maximum diameters of > 15 mm and/or height > 5 mm, 10.1 to 14.9 mm and/or height 2.1 to 4.9 mm, and < 10 mm and/or heights of < 2 mm, respectively. By these criteria, 21 patients had large tumors, 14 patients had intermediate tumors, and 17 had small tumors.

Doses were prescribed in terms of Cobalt-rad-equivalents, (CoRE), using an RBE of 1.1 for the proton beam. All patients received five fractions, with total doses ranging from 4600 to 10,000 CoRE delivered over 6 to 15 days. Most (60%) received 7000 CoRE, and 71% completed treatment in 7 or 8 days.

Treatment was given through individualized brass apertures using a horizontal beam, with patient sitting and immobilized by a head holder or an individualized thermoplastic face mask. A bite block was also employed to aid in immobilization. X-ray localization of the tantalum rings was employed for initial treatment planning and for subsequent patient set-ups, as well as for verification of positioning during treatment. The eye was maintained in the desired position during setup and treatment by voluntary fixation on a specific point chosen to minimize the dose to sensitive ocular structures, especially the lens and the macula. Video-camera monitoring of the front of the eye during treatment ensured that eye motion was less than 0.5 mm in any direction.

Tumor regression has been observed in 29 of the 30 patients (87%) with a minimum 12 month follow-up. The only patient not showing regression after treatment had the largest tumor (17 mm x 17 mm x 12 mm), and developed com-